

Executive Summary

Euro-Office with Nextcloud Air-Gapped Deployment for Secure Document Management

STATUS	NAME	ENVIRONMENT	IMAGE	UPTIME	ACTIONS
Running	euro-office	podman	ghcr.io/euro-office/documentserver:latest	1 day	[Stop] [Restart] [Logs]
Running	nextcloud-ai (compose)	podman			[Stop] [Restart] [Logs]
Running	mastercontainer	podman	ghcr.io/nextcloud-releases/ai-in-one:latest	29 minutes	[Stop] [Restart] [Logs]
Exited	domaincheck	podman	ghcr.io/nextcloud-releases/ai-domaincheck:latest		[Stop] [Restart] [Logs]
Running	eurooffice	podman	ghcr.io/nextcloud-releases/ai-eurooffice:latest	52 minutes	[Stop] [Restart] [Logs]
Running	talk	podman	ghcr.io/nextcloud-releases/ai-talk:latest	52 minutes	[Stop] [Restart] [Logs]
Running	notifypush	podman	ghcr.io/nextcloud-releases/ai-notifypush:latest	52 minutes	[Stop] [Restart] [Logs]
Running	whiteboard	podman	ghcr.io/nextcloud-releases/ai-whiteboard:latest	51 minutes	[Stop] [Restart] [Logs]
Running	database	podman	ghcr.io/nextcloud-releases/ai-postgresql:latest	51 minutes	[Stop] [Restart] [Logs]
Running	redis	podman	ghcr.io/nextcloud-releases/ai-redis:latest	51 minutes	[Stop] [Restart] [Logs]
Running	imaginary	podman	ghcr.io/nextcloud-releases/ai-imaginary:latest	51 minutes	[Stop] [Restart] [Logs]
Running	nextcloud	podman	ghcr.io/nextcloud-releases/ai-nextcloud:latest	26 minutes	[Stop] [Restart] [Logs]
Running	apache	podman	ghcr.io/nextcloud-releases/ai-apache:latest	28 minutes	[Stop] [Restart] [Logs]

Overview: Nextcloud with Euro-Office running as local containers

Classification: Internal / For Decision-Makers

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1. Why We Need an Air-Gapped Environment

Key concept: An "air-gapped" environment means our system has *no connection to the internet*. All data stays within our company network, making it impossible for information to leak outward.

1.1 The Threat Landscape

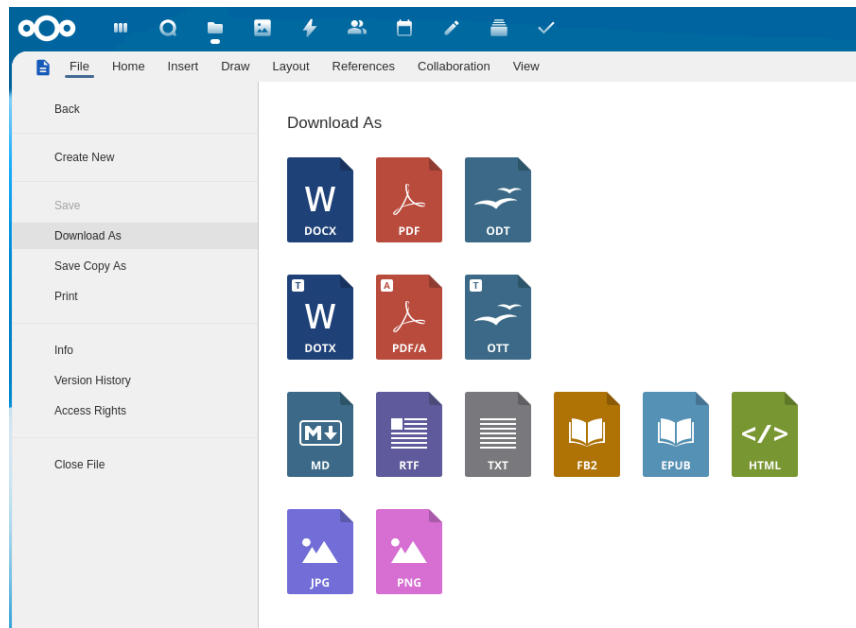
In today's digital environment, organisations face growing risks from:

- **Spy software (spyware):** Malicious programs that secretly monitor user activity, capture keystrokes, and transmit sensitive data to external servers. These threats target documents, emails, passwords, and internal communications.
- **Malicious software (malware):** Ransomware, viruses, and trojans that can encrypt, steal, or destroy company data. A single breach can cost millions in recovery and lost business.
- **Backdoors:** Hidden access points deliberately built into software (or introduced through compromised updates) that allow unauthorised third parties to bypass security measures and access our systems at will.

1.2 How Air-Gapping Eliminates These Risks

By completely severing the connection between our office system and the internet, we achieve:

- **No data leakage:** Even if malware were present in the software, it has no path to transmit stolen data outside the building. Information physically cannot leave.
- **No remote access:** Without an internet connection, no external actor - whether a hacker, intelligence service, or malicious insider using remote tools - can reach our systems.
- **No forced updates:** Software updates (which could potentially introduce spyware or backdoors) are only installed after thorough vetting on a separate, isolated machine before being transferred manually.
- **No telemetry or tracking:** Modern software often sends usage data back to the vendor. In an air-gapped environment, none of this communication is possible.



Euro-Office supports all major document formats: DOCX, XLSX, PPTX, and ODF

1.3 Analogy for Non-Technical Readers

Think of it this way: our current office software (Microsoft Office, Google Docs, etc.) is like having a window in our office that anyone can peek through. An air-gapped system is like putting that office inside a vault with no windows, no phone lines, and no way to send or receive messages - except through a controlled, manual process we fully manage ourselves.

2. What is Euro-Office and Why is it More Trustworthy?

2.1 The Problem with Existing Office Suites

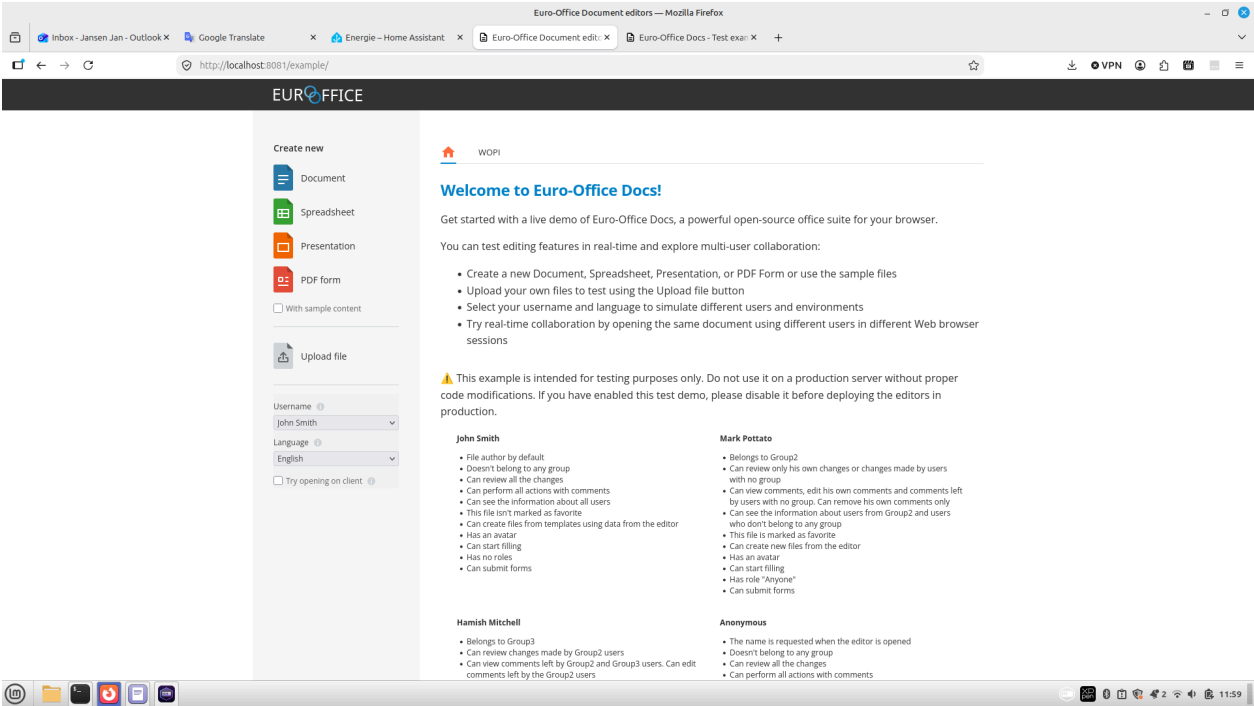
Commercial office software from major vendors contains several features that present security risks for government agencies and organisations handling sensitive data:

- **Cloud-connected features** that require internet access to function fully
- **Telemetry and data collection** that sends usage information to the vendor
- **Third-party integrations** (AI assistants, collaboration plugins) that may connect to external services
- **Proprietary, closed-source code** that cannot be independently audited for backdoors
- **Licensing dependencies** that give the vendor ongoing access and control

2.2 What Makes Euro-Office Different

Euro-Office is an office suite specifically designed for government agencies and security-conscious organisations. It is:

- **Stripped of suspect software:** All unnecessary linkages, cloud connections, AI features, and third-party integrations found in other office suites have been removed.
- **Open source:** The code can be independently verified - there are no hidden backdoors because anyone can inspect the source code.
- **Self-hosted:** The document server runs entirely on our own hardware, inside our network. No data ever leaves the building.
- **Full-featured:** Despite being stripped of risky features, it still supports all standard document formats (Word, Excel, PowerPoint, ODF) with real-time collaborative editing.



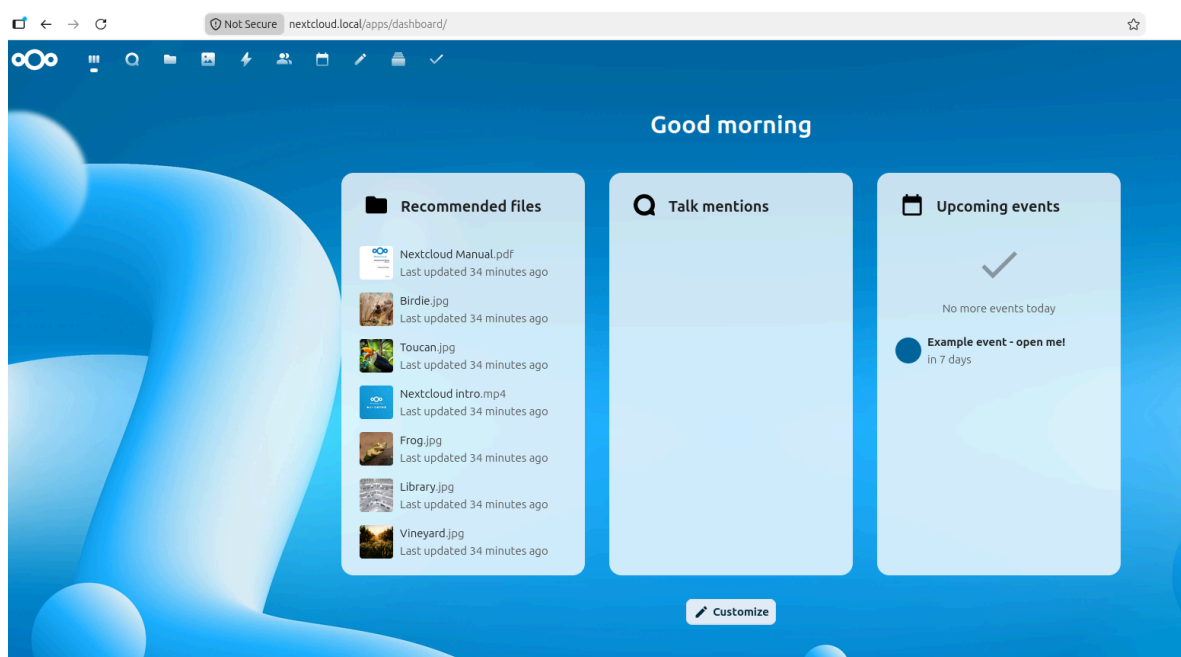
Euro-Office in action: editing a document within the Nextcloud web interface

3. How Nextcloud and Euro-Office Work Together

3.1 Nextcloud: Our File Server

Nextcloud is the **storage and access layer** - think of it as our own private cloud storage, similar to Dropbox or Google Drive, but running entirely on our own hardware. It provides:

- **File storage and organisation:** A web-based interface where users store, share, and organise their documents.
- **User management:** Admins create accounts, set permissions, and control who can access what.
- **Version history:** Every change to a document is tracked and can be reverted.
- **Access from any device on the network:** Users open a browser, navigate to our Nextcloud address, and log in.



The Nextcloud login screen - accessible to all users on the company network

3.2 Euro-Office: The Document Editor

Euro-Office provides the **editing capability**. When a user clicks on a Word document, spreadsheet, or presentation in Nextcloud, Euro-Office opens it in the browser for editing - no installation needed on the user's computer.

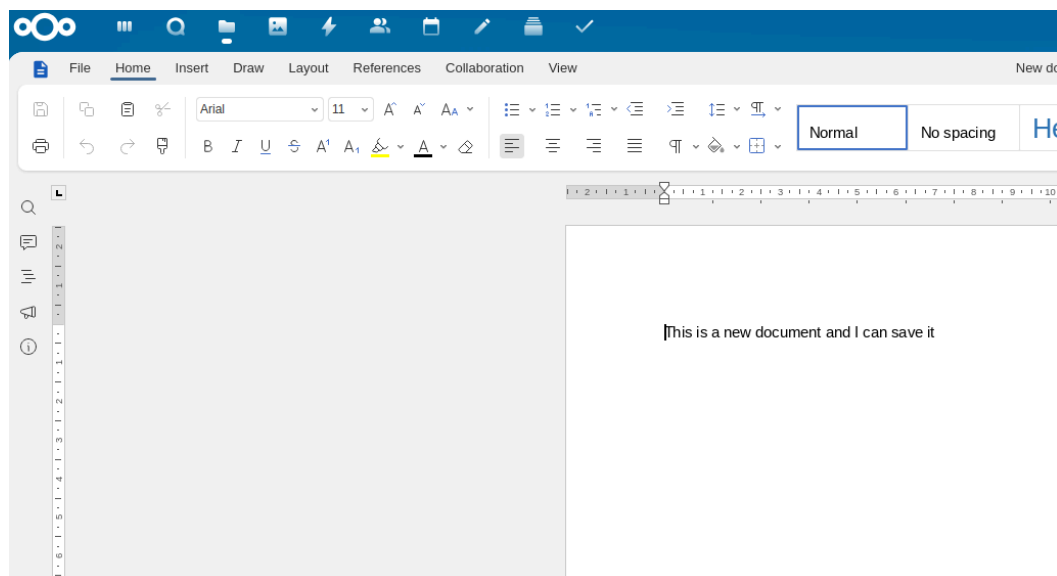
3.3 The Connection Between Them

How it works in practice:

1. A user opens their browser and navigates to our Nextcloud address.
2. They log in and see their files and folders.
3. They click on a document (e.g., a Word file).
4. Euro-Office opens the document in the browser for editing.
5. When the user clicks "Save," the document is stored back in Nextcloud.

All of this happens entirely within our network. No data ever leaves the building.

The two systems communicate through a secure internal connection. Euro-Office knows where to find Nextcloud (to save documents), and Nextcloud knows where to find Euro-Office (to open documents for editing). This connection is configured once during setup and uses a shared secret key (JWT token) to ensure only our systems can communicate with each other.



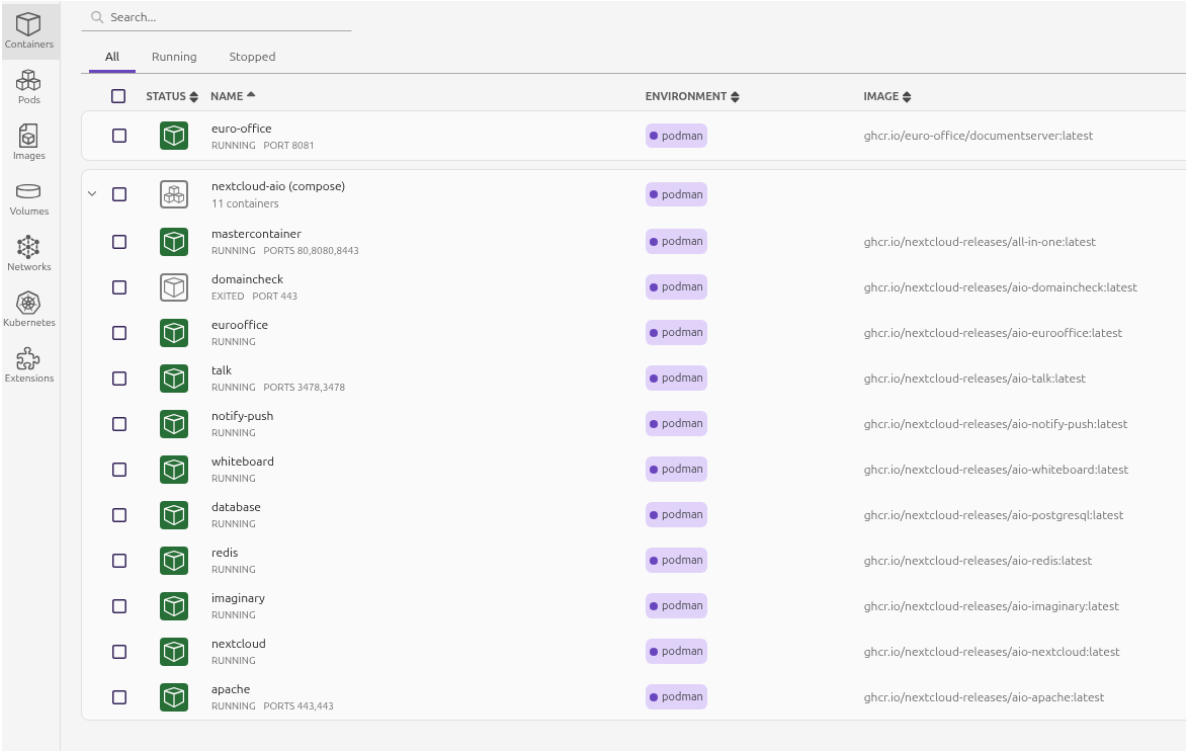
Successfully opening and saving a document through the integrated system

4. Our Installation: What We Did and What We Encountered

4.1 What We Set Out to Do

We deployed three software components as **containers** (think of containers as self-contained packages that include everything the software needs to run, without interfering with the host computer):

- 1. **Nextcloud** - the file server and web interface
- 2. **PostgreSQL** - the database that stores Nextcloud's information (user accounts, settings, file metadata)
- 3. **Euro-Office Document Server** - the editing engine that opens and saves documents



The screenshot shows a container management interface with a sidebar on the left containing icons for Containers, Pods, Images, Volumes, Networks, Kubernetes, and Extensions. The main area has a search bar and tabs for All, Running, and Stopped. A table lists containers with columns for STATUS, NAME, ENVIRONMENT, and IMAGE. The containers listed are: euro-office (RUNNING, PORT 8081, podman, ghcr.io/euro-office/documentserver:latest), nextcloud-aio (compose) (11 containers), mastercontainer (RUNNING, PORTS 80,8080,8443, podman, ghcr.io/nextcloud-releases/all-in-one:latest), domaincheck (EXITED, PORT 443, podman, ghcr.io/nextcloud-releases/aio-domaincheck:latest), eurooffice (RUNNING, podman, ghcr.io/nextcloud-releases/aio-eurooffice:latest), talk (RUNNING, PORTS 3478,3478, podman, ghcr.io/nextcloud-releases/aio-talk:latest), notify-push (RUNNING, podman, ghcr.io/nextcloud-releases/aio-notify-push:latest), whiteboard (RUNNING, podman, ghcr.io/nextcloud-releases/aio-whiteboard:latest), database (RUNNING, podman, ghcr.io/nextcloud-releases/aio-postgresql:latest), redis (RUNNING, podman, ghcr.io/nextcloud-releases/aio-redis:latest), imaginary (RUNNING, podman, ghcr.io/nextcloud-releases/aio-imaginary:latest), nextcloud (RUNNING, podman, ghcr.io/nextcloud-releases/aio-nextcloud:latest), and apache (RUNNING, PORTS 443,443, podman, ghcr.io/nextcloud-releases/aio-apache:latest).

STATUS	NAME	ENVIRONMENT	IMAGE
☐	euro-office RUNNING PORT 8081	podman	ghcr.io/euro-office/documentserver:latest
☒	nextcloud-aio (compose) 11 containers	podman	
☐	mastercontainer RUNNING PORTS 80,8080,8443	podman	ghcr.io/nextcloud-releases/all-in-one:latest
☐	domaincheck EXITED PORT 443	podman	ghcr.io/nextcloud-releases/aio-domaincheck:latest
☐	eurooffice RUNNING	podman	ghcr.io/nextcloud-releases/aio-eurooffice:latest
☐	talk RUNNING PORTS 3478,3478	podman	ghcr.io/nextcloud-releases/aio-talk:latest
☐	notify-push RUNNING	podman	ghcr.io/nextcloud-releases/aio-notify-push:latest
☐	whiteboard RUNNING	podman	ghcr.io/nextcloud-releases/aio-whiteboard:latest
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☐	nextcloud RUNNING	podman	ghcr.io/nextcloud-releases/aio-nextcloud:latest
☐	apache RUNNING PORTS 443,443	podman	ghcr.io/nextcloud-releases/aio-apache:latest

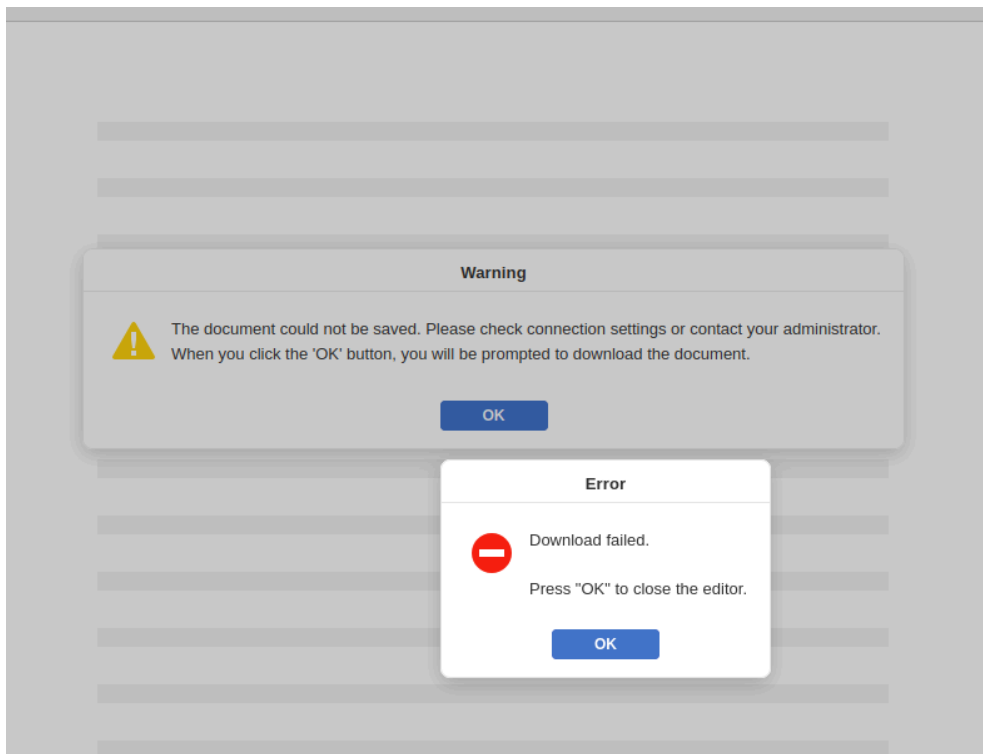
The Euro-Office container running on our server

4.2 Difficulties We Encountered and How We Solved Them

During installation, we encountered several technical challenges. In simple terms, here is what happened and how we resolved each issue:

Problem	What Happened (Simplified)	Our Solution
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Software version mismatch	The Nextcloud management tool expected a newer version of the container engine than what we had installed. The tools could not communicate properly.	We told the software to use the older, compatible version number, which resolved the communication issue.
Permission denied for secure connection	Our container system runs everything as a regular user (not as administrator) for security. But certain components needed permission to create secure web connections (HTTPS) on the standard port. The system blocked this.	We explicitly granted the needed permission to only the specific components that required it, maintaining the overall security principle.
SSL certificate failure	The software tried to obtain an internet-trusted security certificate for our local address. Since our system has no internet connection and uses a local-only name, this could never succeed. The browser refused to connect.	We configured the system to use its own self-signed certificate instead. This works identically for local use - users see a browser warning they can safely accept.
Documents could not be saved	Euro-Office could open documents but could not save them back to Nextcloud. The editor's background processes could not verify the security certificate of our Nextcloud server (because it was self-signed) and refused to communicate.	We told Euro-Office to trust our local server's certificate and to accept connections to private network addresses. This is safe because the entire system is air-gapped.
Connection blocked after restart	After restarting a component, it received a new internal address. Nextcloud only accepted connections from the old address and rejected the new one with "Connection disallowed."	We restarted Nextcloud so it would learn the new address and accept connections again.
Missing software components	The Nextcloud container was missing several required software libraries (PHP extensions) that the Euro-Office integration needed.	We installed the missing libraries inside the container.



An example of an error we encountered: Nextcloud blocking connections after a restart

Important note: A helper script was created that automatically applies all fixes. When the main management container is recreated (e.g., during an update), this script must be re-run to re-apply the fixes. This is an automated process that runs at every session start.

5. Why We Chose Podman Over Docker

For running our software containers, we deliberately chose **Podman** instead of the more widely known Docker. Here is why this matters for security:

5.1 Security Comparison

Feature	Docker	Podman (Our Choice)
Runs as administrator?	Yes - requires a central service running with full system privileges (root access)	No - runs entirely as a regular, unprivileged user by default
Single point of failure?	Yes - a central daemon manages all containers; if compromised, all containers are at risk	No - each container runs independently as its own process
Licensing cost?	Paid subscriptions required for organisations with 250+ employees	Free and open source (Apache 2.0 licence)
Government Linux support?	No longer supported by Red Hat	Default container engine in Red Hat Enterprise Linux 8+
Audit trail?	All actions trace back to the central daemon, obscuring which user performed an action	Direct user tracking - each action is traceable to the individual user account

In plain language: Podman is like having individual locks on each office door that can only be opened by the person assigned to that room. Docker is like having one master key held by a single person - if that key is stolen, every room is compromised.

6. Making It Work on the Company Network (DNS)

6.1 The Problem

Our Nextcloud system needs a friendly name that employees can type into their browser - something like `nextcloud.local` or `euro-office.local` rather than a numeric address like `192.168.0.114`.

Normally, the internet's DNS system (the "phone book" of the internet) translates names to addresses. But since we are air-gapped, we cannot use the internet's DNS.

6.2 The Solution: Local DNS Entry

How it works: We add a single entry to our local network's DNS server (or to each user's computer) that maps the name `euro-office` (or `nextcloud.local`) to the server's internal IP address (e.g., `192.168.0.114`).

This is a simple configuration change that takes minutes to implement.

Option A: Company DNS Server (Recommended)

If the company has a local DNS server (most do), a single record is added:

Record Type	Name	Points To
A Record	euro-office	192.168.0.114
A Record	nextcloud.local	192.168.0.114

After this, every employee on the network can simply type `https://euro-office` in their browser to reach the system.

Option B: Individual Computers (For Testing)

For a quick test setup, each user's computer can have a local hosts file entry. This is a one-line change per computer.

Note: With a local-only address, the browser will show a certificate warning ("Connection is not private"). This is expected and can be safely accepted - it simply means the system uses its own security certificate rather than one from an internet authority.

7. Hardware Requirements for a 10-Person Test Setup

7.1 Minimum Recommended Server

For a test deployment serving up to 10 concurrent users, the following hardware is recommended:

Component	Minimum	Recommended	Notes
Processor (CPU)	4 cores	8 cores	The Euro-Office document server is CPU-intensive when converting/rendering documents. More cores = faster document opening.
Memory (RAM)	16 GB	32 GB	Nextcloud needs ~2 GB, PostgreSQL ~2 GB, Euro-Office needs 4-8 GB depending on concurrent users. 32 GB provides comfortable headroom.
Disk Space	250 GB	500 GB SSD	SSD strongly recommended for responsiveness. The containers themselves need ~20 GB; the rest is for document storage.
Network	1 Gbps	1 Gbps	Standard office network. No internet connection required.
Operating System	Linux (Ubuntu 24.04 LTS, Debian 12, or Fedora 41+). The test was conducted on Linux Mint 22.3.		

7.2 Software Stack (Runs on the Server Above)

Component	What It Does	Resource Usage
Podman	Container engine (runs the other components in isolation)	Minimal
Nextcloud	File server and web interface	~2 GB RAM, ~5 GB disk
PostgreSQL	Database for user accounts, settings, file metadata	~1-2 GB RAM, ~5 GB disk
Euro-Office Document Server	Opens and edits documents in the browser	4-8 GB RAM, ~10 GB disk

Summary for procurement: A standard server with an 8-core processor, 32 GB RAM, and a 500 GB SSD running Linux will comfortably serve 10 users in a test environment. No special

hardware is needed - a typical departmental server suffices.

7.3 Estimated Cost Range

For a suitable server (e.g., Dell PowerEdge T350, HP ProLiant ML30, or equivalent):

- **Hardware:** approximately EUR 1,500 - 3,000 depending on configuration
- **Software licensing:** EUR 0 (all components are open source and free)
- **Setup labour:** approximately 1-2 days of IT staff time for initial deployment

8. Next Steps: From Test to Production (Kubernetes)

8.1 The Test Setup (Current: Podman)

Our current test deployment uses **Podman** on a single server. This is ideal for:

- Validating the system works for our use case
- Testing with a small group of users
- Low-cost proof of concept
- Quick setup (hours rather than days)

8.2 The Production Setup (Future: Kubernetes)

When rolling out to the entire organisation, **Kubernetes** is the recommended platform. Kubernetes is an industrial-strength container management system that provides:

- **High availability:** If one server fails, the system automatically continues on another server - no downtime.
- **Scalability:** As the number of users grows, we add more servers to the cluster. The system distributes the load automatically.
- **Automated recovery:** Failed components are automatically restarted and rescheduled.
- **Rolling updates:** Software can be updated without taking the system offline.
- **Centralised management:** All components are managed through a single interface with proper audit trails.

Recommendation: Start with the Podman test setup (10 users, 1 server). If successful, plan a Kubernetes deployment for the full organisation. The same software components are used - only the management layer changes.

9. Conclusion and Recommendations

9.1 Summary of Benefits

Benefit	Impact
Air-gapped (no internet)	Eliminates data leakage, spyware transmission, and remote access risks
Euro-Office (open source, stripped)	Trustworthy office suite without backdoors or hidden connections
Nextcloud (self-hosted)	Complete control over file storage with no third-party dependencies
Podman (rootless, daemonless)	Meets government security standards; zero licensing cost
Local network access	All employees on the company network can access the system via a simple web address

9.2 Recommended Next Steps

- 1. Approve the test deployment** - Procure a server with the specified hardware and deploy the Podman-based test setup for 10 users.
- 2. Configure local DNS** - Add the DNS entries to the company network so employees can access the system by name.
- 3. User acceptance testing** - Have the test group create, edit, and save documents to validate the workflow.
- 4. Evaluate for production** - Based on test results, plan the Kubernetes-based production deployment.
- 5. Full rollout** - Deploy to all employees with proper training and support.

9.3 Key Advantage

This solution gives our organisation a fully functional, modern office suite with real-time collaboration capabilities, running entirely on our own infrastructure with **zero data leakage risk to the internet**. It is a trustworthy alternative to commercial cloud-based office solutions, at a fraction of the ongoing licensing cost.